Concrete Mix Design Annotated Mix Design

Delivered to: Nebraska Department of Transportation

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1. Input Collection Logic

The script begins by gathering the same data points required by the "Mix Design" tab of the NDOT workbook.

- Project Identification: Variables for Project No, Control No, and Concrete Class are captured to ensure the output is professional and traceable.
- Data Types: All numerical inputs are wrapped in float(input()) to ensure the program can process the decimal values required for Water/Cement ratios and Specific Gravities.
- Sequential Order: The input prompts are hardcoded to match the visual flow of the NDOT submittal sheet to minimize user error.

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2. Excel-to-Python Formula Translation

The following table serves as the core of the ACD, mapping the original Excel cell logic directly to the Python code.

Excel Calculation Description	Original Excel Formula	Python Script Implementation
Wt. of Water (Q)	=(A+B+C+D)*E	Q = (A + B + C + D) * E
Volume of Cement (R)	=A/(J*62.4)	R = A / (J * 62.4)
Volume of Fly Ash (S)	=B/(K*62.4)	S = B / (K * 62.4)
Volume of Silica Fume (T)	=C/(L*62.4)	T = C / (L * 62.4)
Volume of Other SCM (U)	=D/(M*62.4)	U = D / (M * 62.4)
Volume of Air (V)	=(F/100)*27	V = (F / 100) * 27
Volume of Water (W)	=Q/62.4	W = Q / 62.4
Vol. of Total Aggregate (X)	=27-R-S-T-U-V-W	X = 27 - R - S - T - U - V - W

Excel Calculation Description	Original Excel Formula	Python Script Implementation
Wt. of Fine Aggregate (Y)	=62.4*(G/100)*N*X	Y = 62.4 * (G / 100) * N * X
Wt. of Coarse Aggregate (Z)	=62.4*(H/100)*O*X	Z = 62.4 * (H / 100) * O * X
Wt. of Other Aggregate (AA)	=62.4*(I/100)*P*X	AA = 62.4 * (I / 100) * P * X

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3. Technical Logic Breakdown

The Python implementation includes specific safeguards and formatting that improve upon the original Excel model:

- Zero-Value Handling: Unlike Excel, which may return #DIV/0! errors, the Python script uses conditional checks (e.g., if J > 0) to handle instances where specific materials (like Silica Fume) are not used in a mix.
- Constant Density: The value 62.4 is treated as a global constant for the unit weight of water (lb/ft^3) to convert mass to volume across all functions.
- Yield Verification: The code calculates the final sum of volumes (R+S+T+U+V+W+X) to verify it equals 27.00 ft^3, confirming the mix satisfies the Absolute Volume Method for exactly one cubic yard.
- Professional Output: The script utilizes f-string alignment (e.g., :<25) to produce a professional weight chart, meeting the client's request for "professional, client-readable format".